# COBOL Banking Practice – Question 33

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## Question

**Transaction Date Validation (Phase 4 – Mainframe Applications | Difficulty: Beginner)**

A banking batch rejects transactions with invalid calendar dates. Read a date and determine whether it is a valid Gregorian calendar date, including leap-year rules. The program must account for month lengths and leap years, and print VALID DATE or INVALID DATE.

Sample Input: 20240229 Sample Output: VALID DATE

## Aim

To write a COBOL program that reads a date, separates it into day, month and year, checks the month and day ranges, applies month-length and leap-year rules, and displays whether the date is valid.

## Procedure

1. Start the program and declare the working-storage variables for the date, day, month, year, a validity flag (T = 1) and the February length (28).
2. Display the prompt and accept the 8-digit date into DMY.
3. Divide DMY by 10000 to get the year as the remainder and the day-month part (DMYC) as the quotient.
4. Divide DMYC by 100 to get the day (quotient) and the month (remainder).
5. If the month is outside 1–12 or the day is outside 1–31, set the flag T to 0.
6. Check the leap-year condition and set the February limit to 29 when it is satisfied.
7. Check month lengths: February days must not exceed the February limit, and 30-day months must not have a day above 30; otherwise set T to 0.
8. If T is 1, display VALID DATE; else display INVALID DATE. Stop the run.

## Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. DATEVALID.
DATA DIVISION.
    WORKING-STORAGE SECTION.
        01 DMY PIC 9(8).
        01 DMYC PIC 9(8).
        01 D PIC 9(2).
        01 M PIC 9(2).
        01 Y PIC 9(4).
        01 T PIC 9(1) VALUE 1.
        01 FEB_L PIC 9(2) VALUE 28.
PROCEDURE DIVISION.
    DISPLAY "Enter a date(YYYYMMDD): "
    ACCEPT DMY
    DIVIDE DMY BY 100 GIVING DMYC REMAINDER D
    DIVIDE DMYC BY 100 GIVING Y REMAINDER M
    IF M>12 OR M<1 OR D<1 OR D>31 THEN
        SET T TO 0
```

```
    END-IF
    IF (FUNCTION REM(Y,4)) IS = 0 AND (FUNCTION REM(Y,400)) IS NOT = 0 THEN
        SET FEB_L TO 29
    END-IF
    IF (FUNCTION REM(M,2) IS = 0) THEN
        IF ((M IS = 2) AND (D > FEB_L)) OR (M<8 AND D>30) THEN
            SET T TO 0
        END-IF
    ELSE
        IF (M>7 AND D>30) THEN
            SET T TO 0
        END-IF
    END-IF
    IF T IS = 1 THEN
    DISPLAY "VALID DATE"
    ELSE
    DISPLAY "INVALID DATE"
    END-IF
STOP RUN.
```

## Output

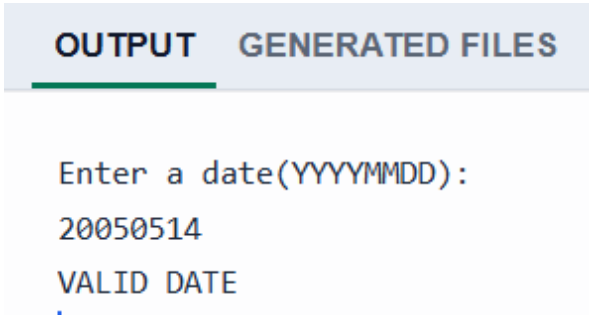



*Fig. 1: Valid date (20050514)*

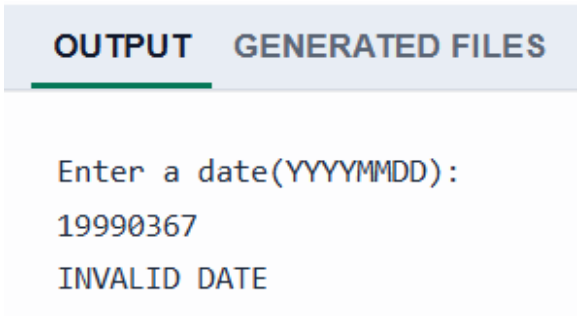



*Fig. 2: Invalid day (19990367)*

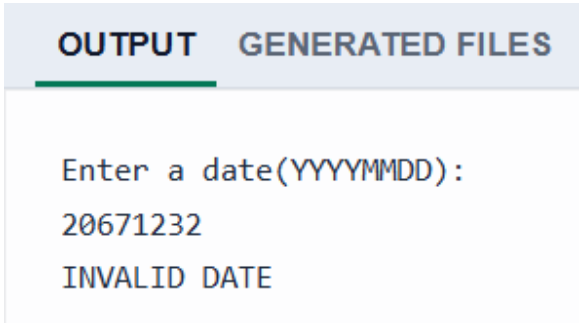



*Fig. 3: Invalid day (20671232)*

## Result

The COBOL program was executed successfully and correctly identified valid and invalid calendar dates. Thus, the date validation program was implemented and verified.